

# The GREA model: description for the CosmoVerse Cosmology Compilation

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We summarise the General Relativistic Entropic Acceleration (GREA) model and its implementation in the **CLASS** Boltzmann code for the CosmoVerse Cosmology Compilation, detailing the modifications to the standard code, the relation to  $\Lambda$ CDM, and the priors adopted on the extra parameter.

## I. GENERAL RELATIVISTIC ENTROPIC ACCELERATION (GREA)

*Short title (figures):* **GREA.**

General Relativistic Entropic Acceleration (GREA) attributes the present accelerated expansion to the entropy growth of the causal (cosmological) horizon, rather than to a cosmological constant or a dark-energy field [1, 2]. In the covariant, out-of-equilibrium thermodynamic formulation of general relativity, this entropy production sources an entropic force that enters the Einstein equations as an effective bulk viscosity with negative pressure, so that  $\Lambda = 0$  throughout [2, 3]. The dynamics is set by a single dimensionless parameter  $\alpha$ , the ratio of the causal-horizon distance to the entropic curvature scale  $k^{-1/2}$ . Because the entropic term scales as  $\sinh(2\tau)$  in the dimensionless time  $\tau$ , it is negligible at early times and relevant only at low redshift, shifting the onset of acceleration and easing the coincidence problem [3]. The effective equation of state  $w(a)$  evolves with redshift and

shows a transient phantom crossing at  $z \lesssim 2$ , in line with earlier background-level analyses [4–6].

We implement GREA in **CLASS** [7] (v3.3.4) as a new case of the fluid equation-of-state module (`fluid_equation_of_state=GREA`). The entropic sector replaces the cosmological constant, with its density fixed by spatial flatness,  $\Omega_{\text{GREA}} = 1 - \Omega_m - \Omega_r$ ; the tabulated  $w(a)$  is interpolated at run time, the Friedmann equation reads  $H^2 = \rho_{\text{tot}}$  with no explicit  $\Omega_k a^{-2}$  term, and perturbations use the parametrised post-Friedmann (PPF) prescription [8] (`use_ppf=yes`,  $c_s^2 = 1$ ) to cross the phantom divide. All other sectors match  $\Lambda$ CDM. GREA is not nested within  $\Lambda$ CDM: no parameter value reproduces it exactly, and as  $\alpha \rightarrow 0$  the entropic term vanishes, leaving a decelerating, matter-dominated ( $\Lambda = 0$ ) universe rather than flat  $\Lambda$ CDM. We sample the dimensionless combination  $\sqrt{k} \eta_0$  (`sqrt_k_eta0`), a monotonic proxy for  $\alpha$ , with a flat prior  $\sqrt{k} \eta_0 \in [2.5, 4.5]$ ;  $\alpha$  is recovered by inversion. The standard  $\Lambda$ CDM parameters keep their usual priors and no GREA nuisance parameters are added.

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